

The calculation for kurtosis involves finding the average of deviations from the mean raised to the fourth power and then standardizing that average by dividing by the standard deviation raised to the fourth power. A normal distribution has kurtosis of 3.0, so a fat-tailed distribution has a kurtosis above 3.0 and a thin-tailed distribution has a kurtosis below 3.0.

Excess kurtosis is the kurtosis relative to the normal distribution. For a large sample size ($n = 100$ or more), **sample excess kurtosis** (K_E) is approximately as follows:

$$K_E \approx \left[\left(\frac{1}{n} \right) \frac{\sum_{i=1}^n (X_i - \bar{X})^4}{s^4} \right] - 3. \quad (9)$$

As with skewness, this measure is free of scale. Many statistical packages report estimates of sample excess kurtosis, labeling this as simply “kurtosis.”

Excess kurtosis thus characterizes kurtosis relative to the normal distribution. A normal distribution has excess kurtosis equal to 0. A fat-tailed distribution has excess kurtosis greater than 0, and a thin-tailed distribution has excess kurtosis less than 0. A return distribution with positive excess kurtosis—a fat-tailed return distribution—has more frequent extremely large deviations from the mean than a normal distribution.

Exhibit 19: Summary of Kurtosis

If kurtosis is ...	then excess kurtosis is ...	Therefore, the distribution is ...	And we refer to the distribution as being ...
above 3.0	above 0	fatter-tailed than the normal distribution.	fat-tailed (leptokurtic)
equal to 3.0	equal to 0	similar in tails to the normal distribution.	mesokurtic
less than 3.0	less than 0	thinner-tailed than the normal distribution.	thin-tailed (platykurtic)

Most equity return series have been found to be fat-tailed. If a return distribution is fat-tailed and we use statistical models that do not account for that distribution, then we will underestimate the likelihood of very bad or very good outcomes. Example 6 revisits the EAA Equity Index from the earlier Example 1 and quantifies the shape of its return distribution.

EXAMPLE 6

Skewness and Kurtosis of EAA Equity Index Daily Returns

Consider the statistics in Exhibit 20 for the EAA Equity Index:

Exhibit 20: Properties of Skewed Distributions

	Daily Return (%)
Arithmetic mean	0.0347
Standard deviation	0.8341
	Measure of Symmetry
Skewness	−0.4260
Excess kurtosis	3.7962